Question - 01

1. What are threads ? Explain in your own words. Why do we need threads ? Explain real world applications.
2. What is the difference between Runnable interface and Threads class ?
3. Write a simple application in java to elaborate movie ticket booking and how threads can be in place to assure the effectiveness in that process. Writing the threads class`s operation is enough.

Question – 02

1. What are the different types of Design Patterns ?
2. What is the relationship of an architecture and a design pattern ?
3. Explain the total operation of Singleton pattern using a Pseudocode.
4. Convert the above Pseudocode into a simple Java Program.

Question – 03

1. Propose a suitable use case diagram for a vehicle service station. Having 05 use cases will be enough.
2. Write use case specifications for the specified use cases above.
3. Explain the process of converting a use case operation to a sequence diagram.
4. Draw a sequence diagram for a desired use case which you have recognized above.

Question – 04

1. What is the difference between get and post methods ?
2. What is a Web Server ? Is it a software or hardware ? Elaborate your ideology.
3. How servlets are different from jsp pages ?
4. Write java code to implement a simple user registration system via using jsp pages and servlets at your discretion .